

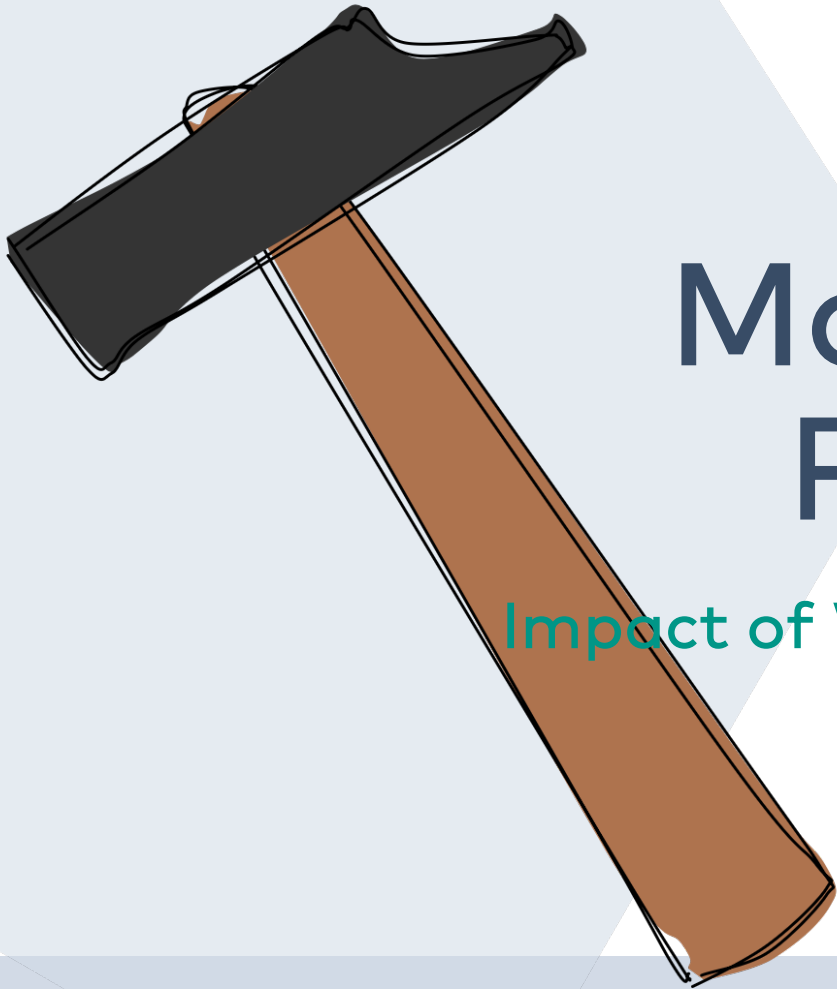
# XLC

Crossload Consulting



## Mastering HANA Performance:

Impact of Workload Management & Partitioning



# XLC

## Crossload Consulting



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# Agenda:

## 1. Workload Management

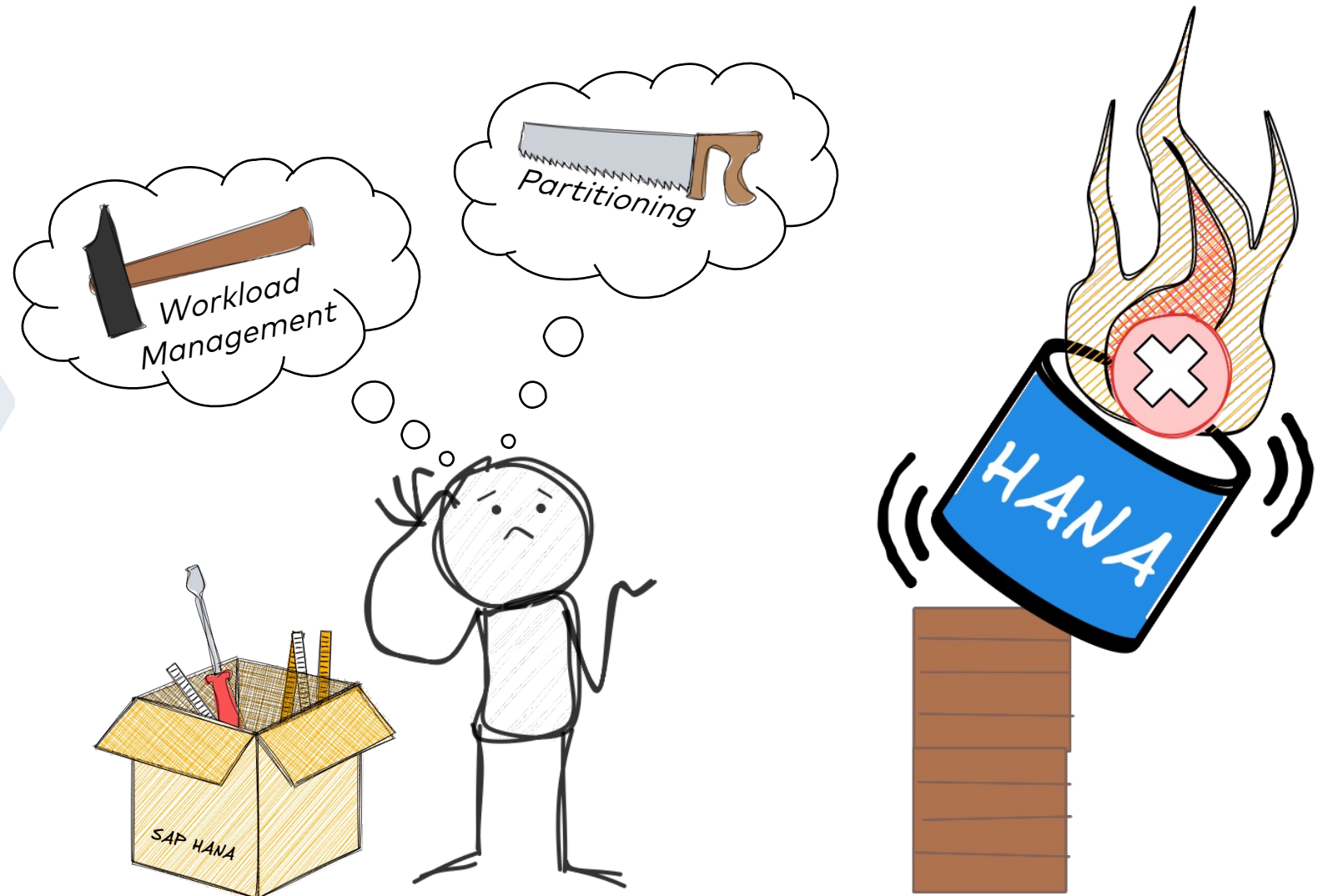
- Overview
- Order
- Workload Classes
- Admission Control
- Hints

## 2. Partitioning

- Overview
- Performance
- Stability

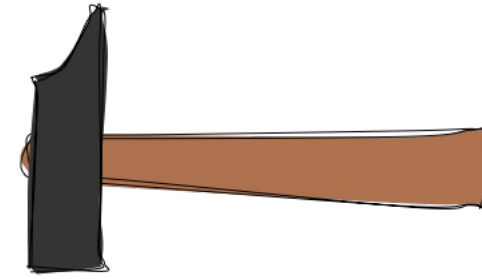
## 3. Combined Use Cases

## 4. Summary



# Overview

## Workload Management

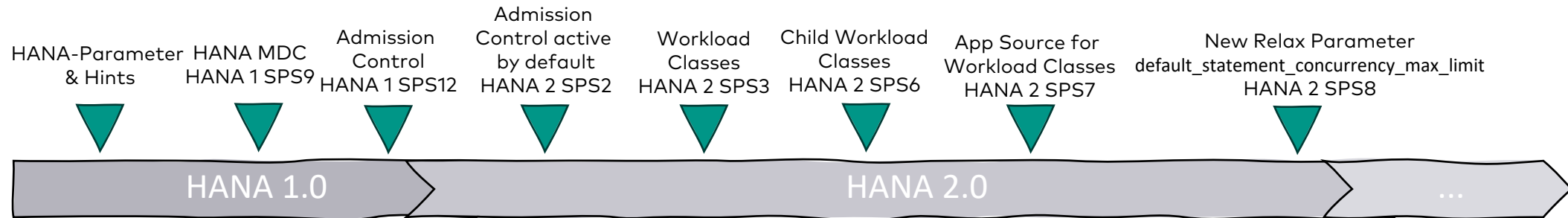


- The objective behind Workload Management is to avoid resource bottlenecks by controlling how these limited resources are used. It consists of a lot of different features

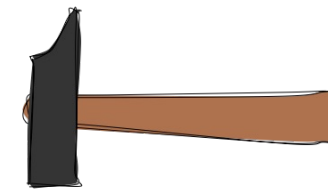
- HANA-Parameters
- Admission Control
- Workload classes
- Hints



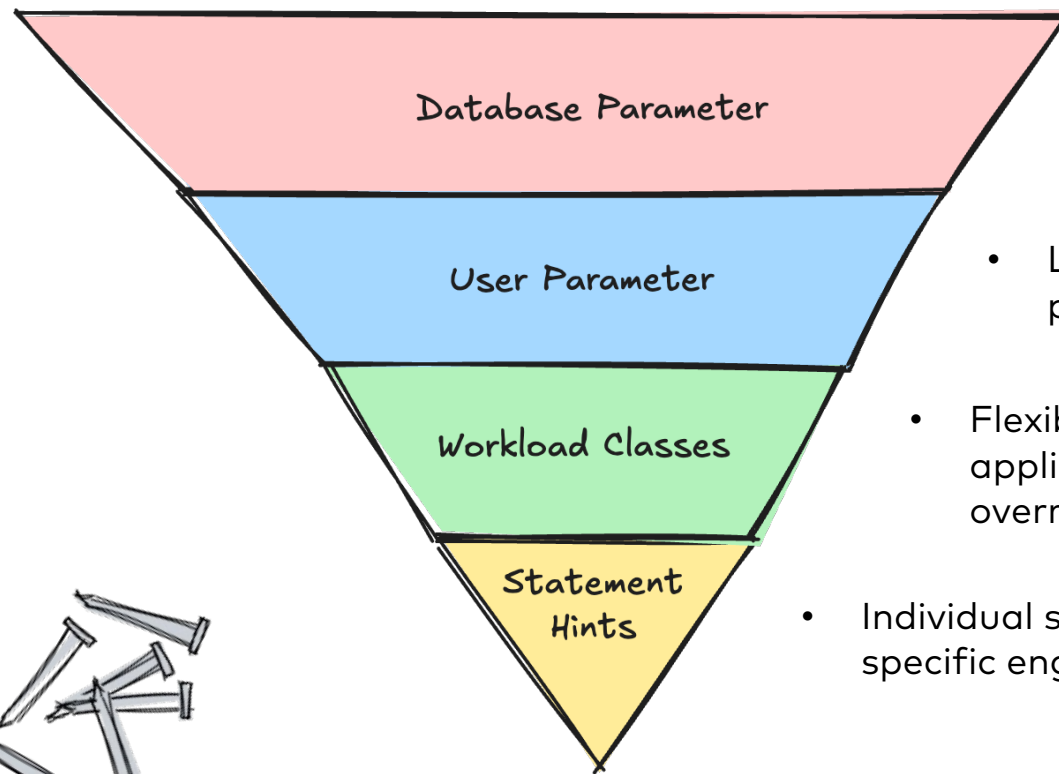
- Over the last SPS new features and functionalities were added as well as steady improved



# Workload Management



## Order of Workload Settings



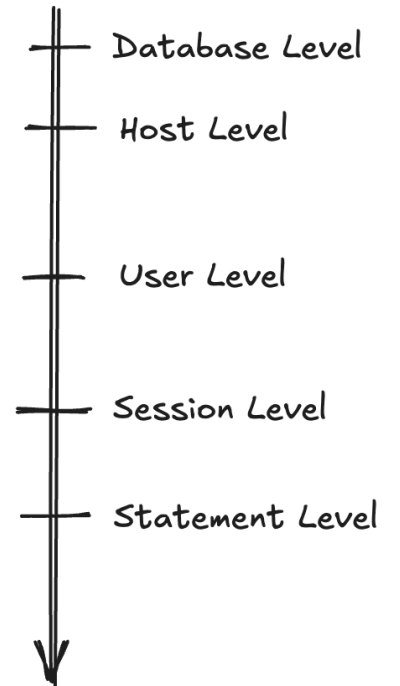
- General Parameters for example to avoid resource usage by one single request

- Limitation on user level to overrule database parameters for single DB users. Hardly ever used.

- Flexible way to control for example complete applications, (but not all parameters can be overruled like `total_statement_memory_limit`)

- Individual statements can be adjusted to use for example specific engines to fine tune performance

very generic



More granular, very specific



# Workload Management

## Workload Classes

### What are Workload Classes?

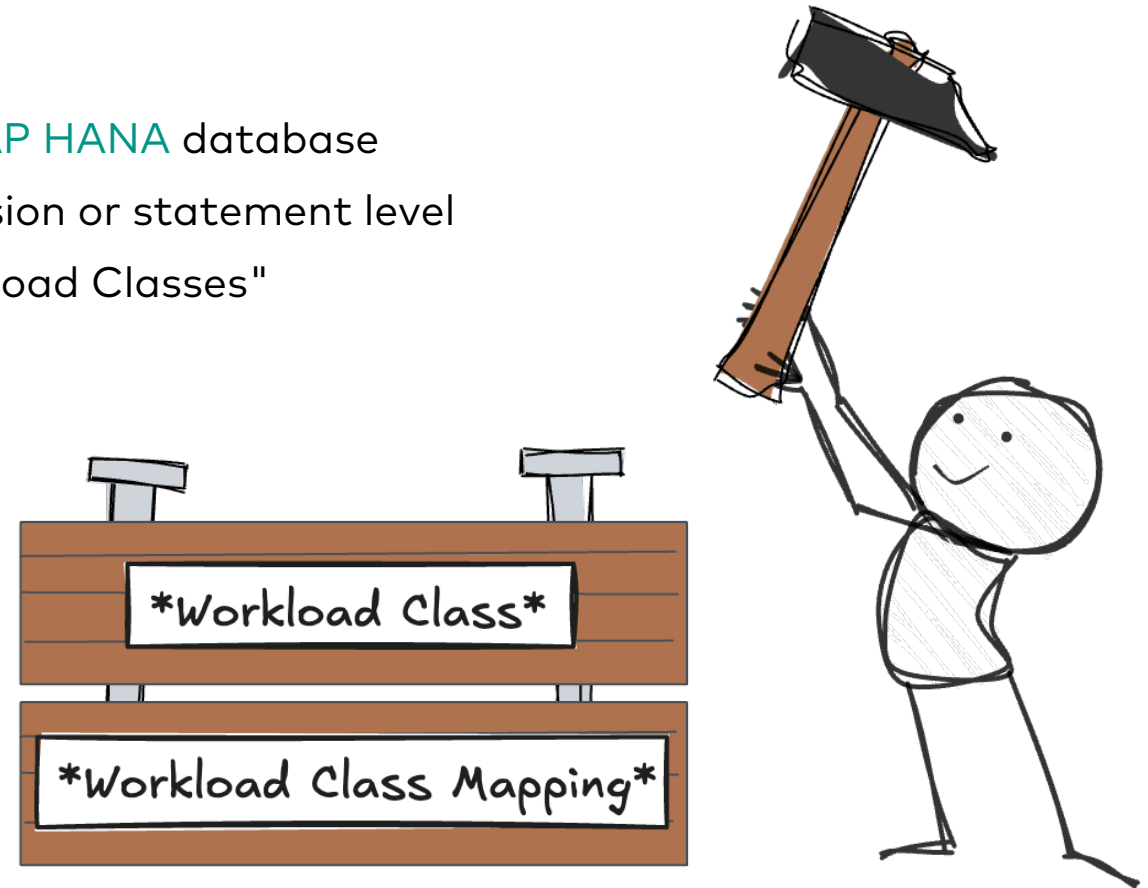
- Possibility to manage **different workloads in one SAP HANA** database
- Allows **dynamic resource management** on user, session or statement level
- Can be created in HANA Cockpit → "Manage Workload Classes"

or

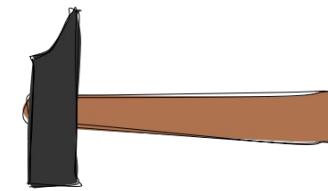
via SQL :

```
CREATE WORKLOAD CLASS "ABC" SET  
  'TOTAL STATEMENT MEMORY LIMIT' = '1000',  
  'TOTAL THREAD LIMIT' = '100';
```

```
CREATE WORKLOAD MAPPING "BI_APP_ABC"  
  WORKLOAD CLASS "ABC"  
  SET 'USER NAME' = 'APP_USER';
```



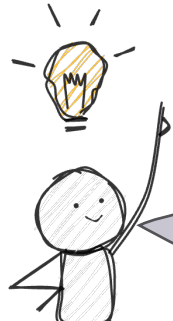
# Workload Management



## Workload Class Mappings

Only with a **valid** mapping Workload Classes are used!

- This mappings need to be created individually
- There can only be **1:1** mapping
- If multiple mappings would fit, they are **prioritized** on
  - a. the number of used session variables
  - b. if the number is identical in this order:
    1. APPLICATION USER NAME
    2. CLIENT
    3. APPLICATION COMPONENT NAME
    4. APPLICATION COMPONENT TYPE
    5. APPLICATION NAME
    6. USER NAME



→ In the Context of Workload Classes and Mappings  
a Naming concept is recommended

### Session Variables

Application Source

Equals

XS Advanced Application User Name

Equals

Application User Name

Equals BIUSER

Client

Equals

Application Component Name

Equals

Application Component Type

Equals

Application Name

Equals



### Session Variables

Application Source

Equals

XS Advanced Application User Name

Equals

Application User Name

Equals BIUSER

Client

Equals 100

Application Component Name

Begins With BI\_APP1

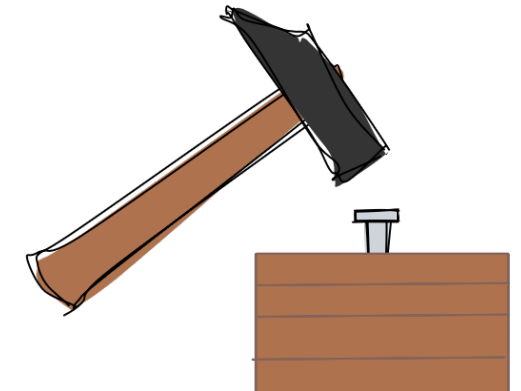
Wildcard: %

Application Component Type

Equals

Application Name

Equals

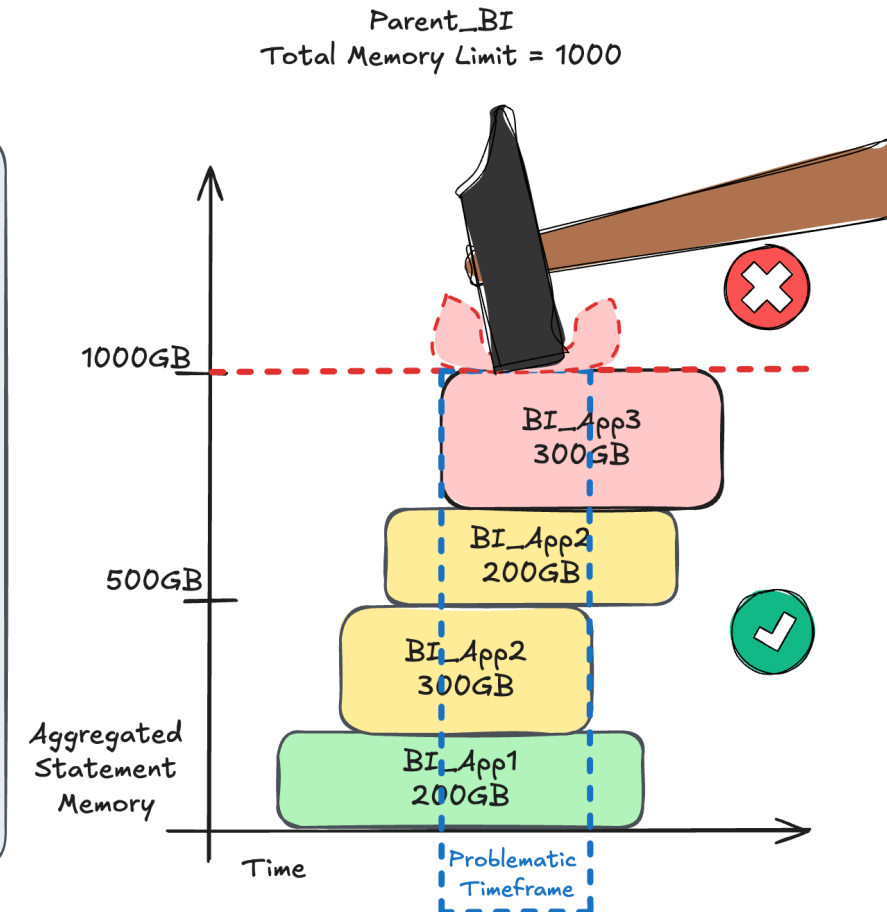
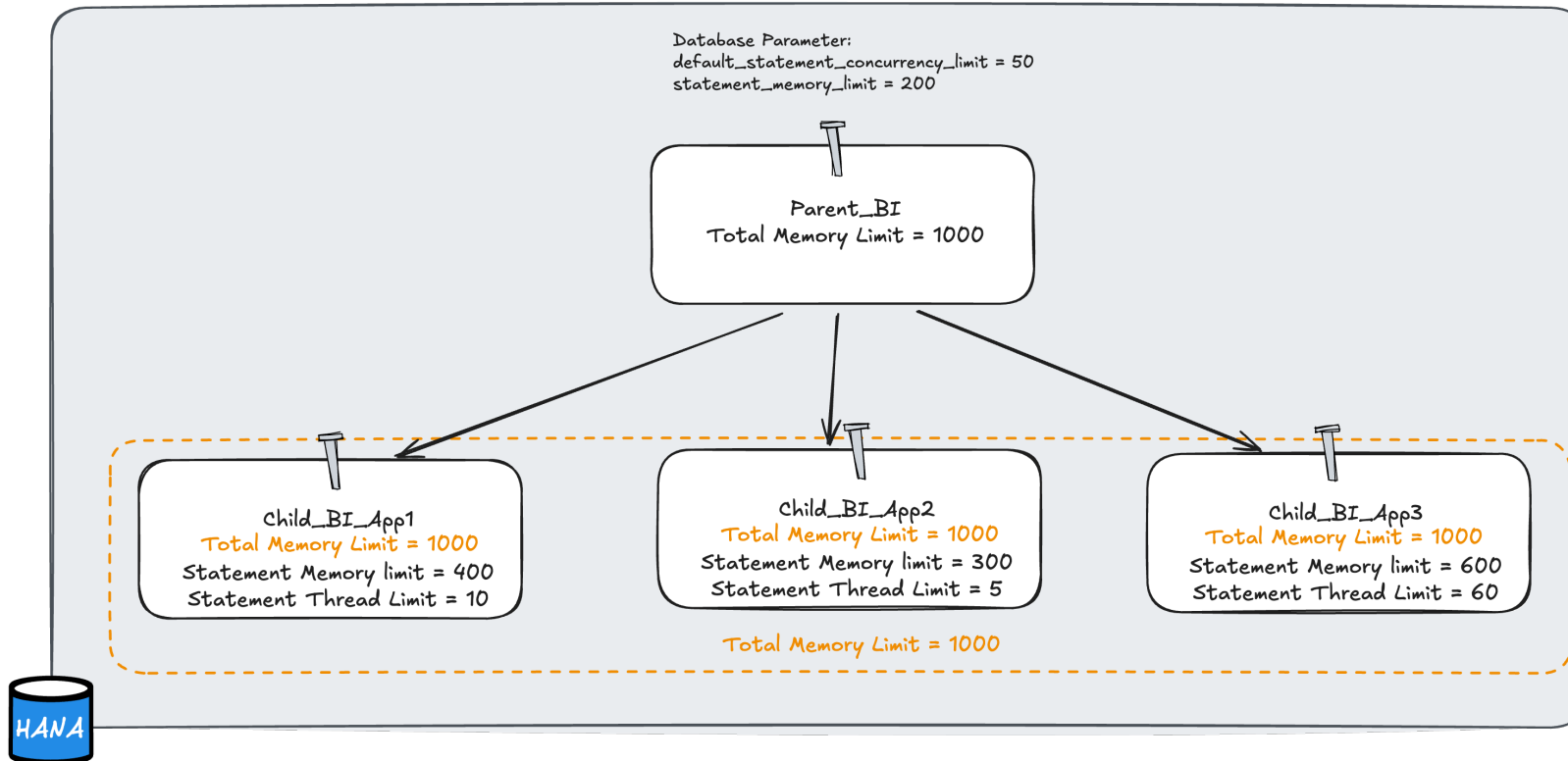




# Workload Management

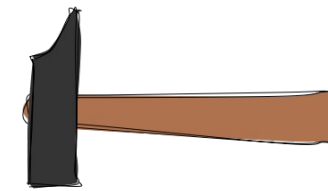
## Workload Class Hierarchies

- Are an interesting approach to limit the overall resource consumption of multiple applications by a parent class

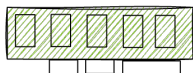




# Workload Management

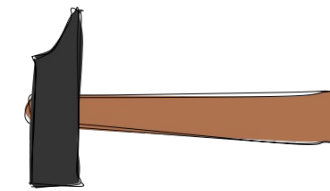


## ➤ Admission Control (AC)

- Is used to **prevent** the HANA from **overload** situations
- Instead, the requests get **queued** or **rejected**
- The thresholds can be defined on **CPU**  or **Memory**  level
- Can be used on **global** level or **inside Workload Classes**
- Workload Class AC is **overruling** global AC



# Workload Management



## Admission Control in Workload Classes

### ■ Parent

#### Edit Workload Class

Workload Class Name: \* Parent\_BI  
Execution Priority: Use default value  
Parent: None  
Is Parent: ☒ YES

#### Total Aggregated Statement Limits

Total Statement Memory Limit: 1000 GB  
Total Statement Thread Limit: 100 threads

#### Resource Limits

Query Timeout: seconds  
Uncommitted Write Lifetime Limit: minutes  
Idle Cursor Lifetime Limit: minutes  
Queue CPU Threshold: %  
Queue Memory Threshold: %  
Reject CPU Threshold: %  
Reject Memory Threshold: %

### ■ Child

#### Edit Workload Class

Workload Class Name: \* Child\_BI\_App1  
Execution Priority: Use default value  
Parent: Parent\_BI  
Total Statement Memory Limit: 1000 GB  
Total Statement Thread Limit: 100 threads  
Is Parent: ☐ NO

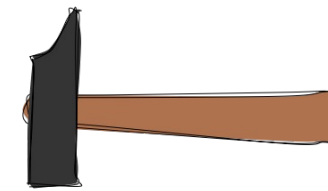
#### Individual Statement Limits

Statement Memory Limit: \* 400 GB  
Statement Thread Limit: \* 10 threads

#### Resource Limits

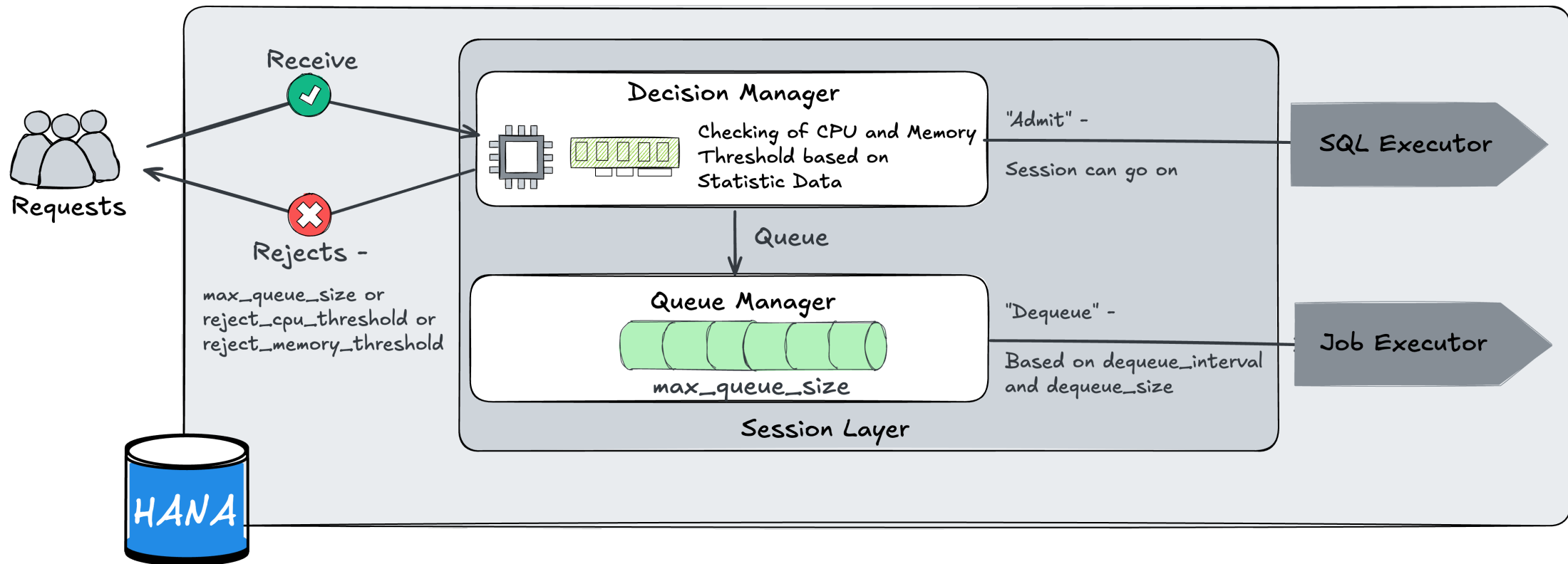
Query Timeout: seconds  
Uncommitted Write Lifetime Limit: minutes  
Idle Cursor Lifetime Limit: minutes  
Queue CPU Threshold: %  
Queue Memory Threshold: %  
Reject CPU Threshold: %  
Reject Memory Threshold: %

# Workload Management



## Admission Control (AC)

- Once a session is permitted by the Decision Manager it is not restricted anymore!



# Workload Management

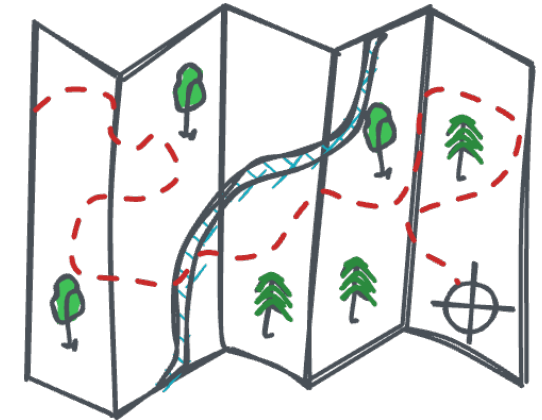
## Database Hints

### What are Hints?

Hints are used to help the SQL Optimizer to find the best way to access data

### What is the difference between SAP HANA Hints and SAP ABAP Hints?

ABAP Hints are part of the ABAP coding and are forwarded to the HANA while HANA Hints are tied to the SQL-HASH or SQL-Text



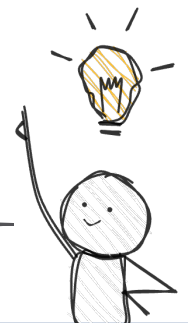
### Which SAP HANA Hints exist?

Not all Hints are documented by SAP but an incomplete list is available in the SAP help documentation here : <https://help.sap.com>

Additional Hint Infos:

[2142945](#) - FAQ: SAP HANA Hints

[2400006](#) - FAQ: SAP HANA Statement Hints



# Workload Management

## Database Hints

In which scenarios are statement hints useful?

- Improve performance
- Reduce resource consumption
- As a workaround for bugs

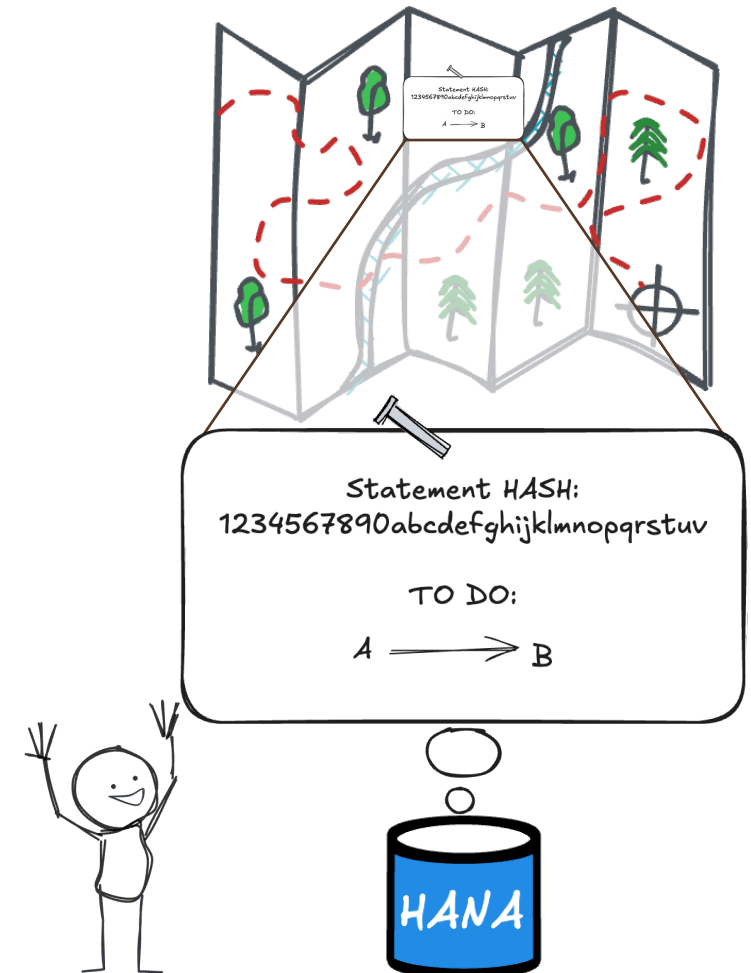
Hints can be set for

- SQL-Hashes:

```
ALTER SYSTEM ADD STATEMENT HINT (NO_CS_JOIN) COMMENT 'Description'  
FOR STATEMENT HASH '1234567890abcdefghijklmnopqrstuvwxyz';
```

- SQL-Text:

```
ALTER SYSTEM ADD STATEMENT HINT (NO_CS_JOIN) COMMENT 'Description'  
FOR SELECT * FROM AAA, BBB WHERE AAA.X = BBB.X;
```



# Workload Management

## Database Hints

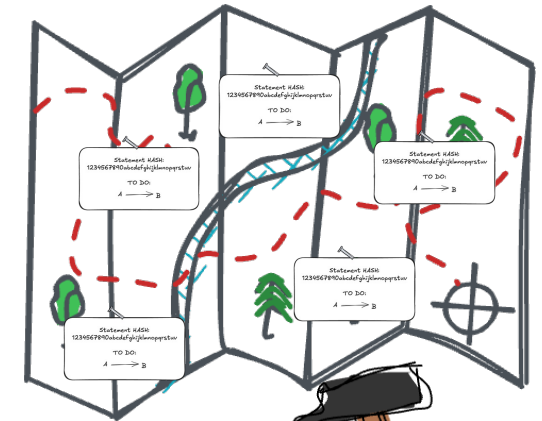
### Attention!

As the HASH changes as soon as the statement text is different statements with IN-lists are difficult to map!

|                                                                     |                                               |
|---------------------------------------------------------------------|-----------------------------------------------|
| <code>SELECT * FROM AAA WHERE B IN ( ? )</code>                     | <code>1234567890abcdefghijklmnopqrstuv</code> |
| <code>SELECT * FROM AAA WHERE B IN ( ? , ? , ? )</code>             | <code>abcdefghijklmnopqrstuv1234567890</code> |
| <code>SELECT * FROM AAA WHERE B IN ( ? , ? , ? , ? , ? , ? )</code> | <code>12345abcdefghijklmnopqrstuv67890</code> |
| ...                                                                 |                                               |

A smart solution for this problem is available starting with SPS08:

```
ALTER SYSTEM ADD STATEMENT HINT (USE_HEX_PLAN) FOR  
SELECT * FROM AAA WHERE B IN ($$LIST$$)
```



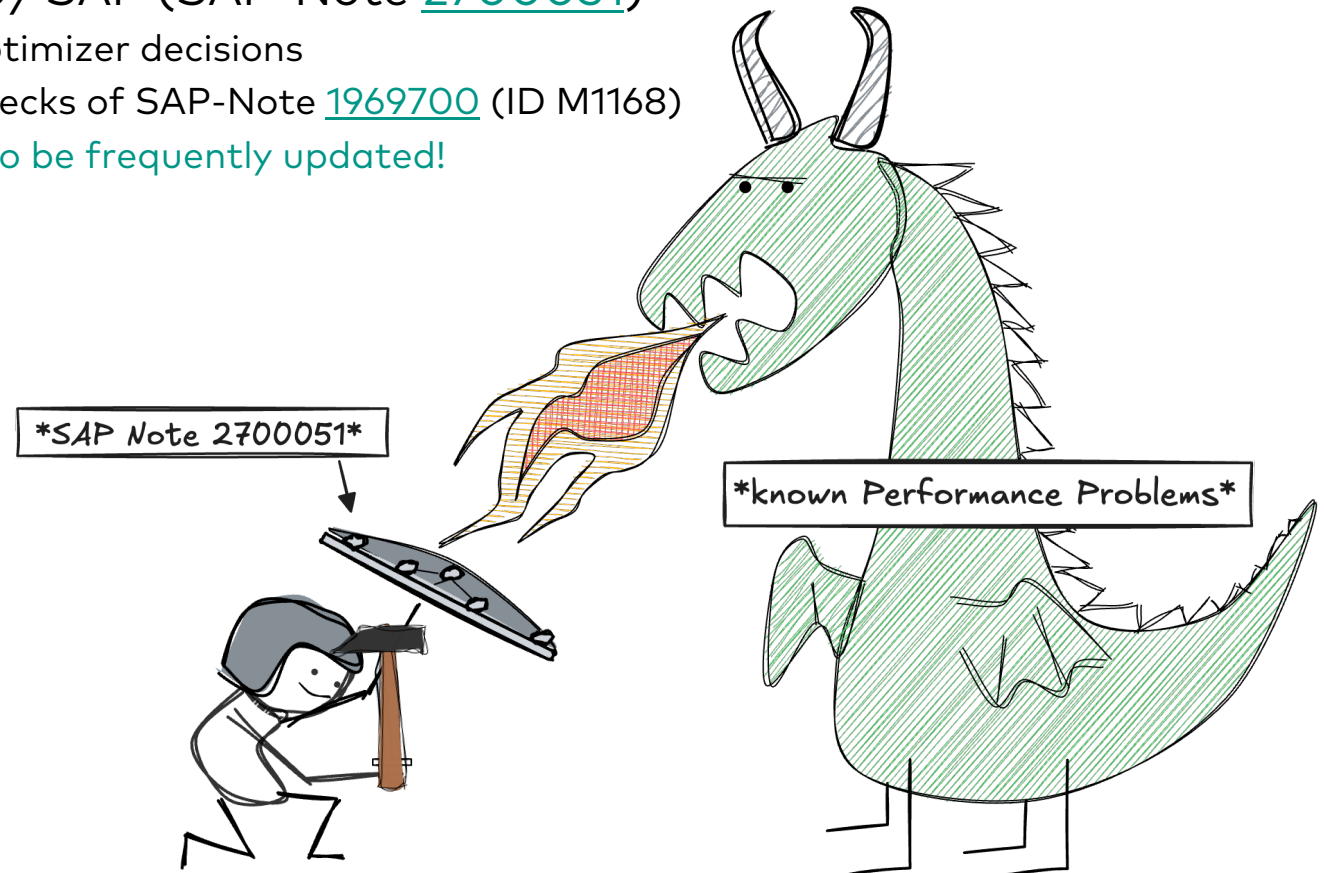
# Workload Management

## Database Hints

- Predefined statement hints available by SAP (SAP-Note [2700051](#))
  - contains hints for known issues with wrong optimizer decisions
  - These are checked in SAP HANA config minichecks of SAP-Note [1969700](#) (ID M1168)
  - Note is frequently updated → **Hints should also be frequently updated!**

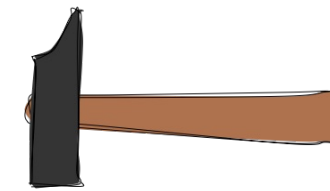
### Recommendation:

Should be implemented and updated every 6 months to avoid Performance Problems





# Workload Management



## Database Hints Example

Script HANA\_SQL\_StatementHash\_DataCollector for statement `zf2f828810342e6318s875f9c289ceb8`:

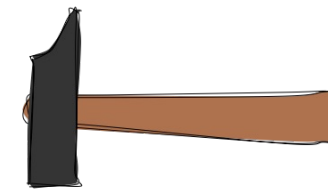
| INSTALL_TIME        | VERSION                |
|---------------------|------------------------|
| 2025/03/17 12:46:21 | 2.00.078.00.1715149848 |
| 2024/02/25 10:58:41 | 2.00.059.11.1700553860 |

| SNP_TIME   | HOST       | PLAN_ID   | ENG | EXECUTIONS | RECORDS | REC_PER_EXEC | ELAPSED_MS | ELA_PER_EXEC_MS | ELA_PER_REC_MS | PREPARES | PREPARE_MS | LOCK_MS |
|------------|------------|-----------|-----|------------|---------|--------------|------------|-----------------|----------------|----------|------------|---------|
| 2025/03/24 | acbdefghij | 161400003 | H   | 155        | 8001    | 51.61        | 473799     | 3056.76         | 59.21          | 2        | 279        | 0       |
| 2025/03/24 | acbdefghij | 866300003 | H   | 1489       | 77128   | 51.79        | 178027     | 119.56          | 2.30           | 0        | 0          | 0       |
| 2025/03/23 | acbdefghij | 866300003 | H   | 1321       | 68366   | 51.75        | 143647     | 108.74          | 2.10           | 0        | 0          | 0       |
| 2025/03/22 | acbdefghij | 866300003 | H   | 21189      | 1084829 | 51.19        | 2469854    | 116.56          | 2.27           | 0        | 0          | 0       |
| 2025/03/21 | acbdefghij | 866300003 | H   | 20473      | 1047748 | 51.17        | 2517974    | 122.99          | 2.40           | 0        | 0          | 0       |
| 2025/03/20 | acbdefghij | 866300003 | H   | 21404      | 1095041 | 51.16        | 2446889    | 114.31          | 2.23           | 0        | 0          | 0       |
| 2025/03/19 | acbdefghij | 866300003 | H   | 9531       | 488898  | 51.29        | 6317029    | 662.78          | 12.92          | 1        | 325        | 0       |
| 2025/03/19 | acbdefghij | 162190003 | H   | 12476      | 639828  | 51.28        | 45630178   | 3657.43         | 71.31          | 0        | 0          | 0       |
| 2025/03/18 | acbdefghij | 162190003 | H   | 19691      | 1009164 | 51.25        | 70485379   | 3579.57         | 69.84          | 0        | 0          | 0       |
| 2025/03/17 | acbdefghij | 162190003 | H   | 20         | 1018    | 50.90        | 62188      | 3109.41         | 61.08          | 2        | 281        | 0       |
| 2025/03/17 | acbdefghij | 107100003 | C   | 779        | 40340   | 51.78        | 38887      | 49.91           | 0.96           | 0        | 0          | 0       |
| 2025/03/16 | acbdefghij | 107100003 | C   | 1075       | 55657   | 51.77        | 61756      | 57.44           | 1.10           | 0        | 0          | 0       |
| 2025/03/15 | acbdefghij | 107100003 | C   | 16197      | 828849  | 51.17        | 846193     | 52.24           | 1.02           | 0        | 0          | 0       |
| 2025/03/14 | acbdefghij | 107100003 | C   | 18911      | 968672  | 51.22        | 959672     | 50.74           | 0.99           | 0        | 0          | 0       |
| 2025/03/13 | acbdefghij | 107100003 | C   | 20043      | 1025885 | 51.18        | 1045158    | 52.14           | 1.01           | 0        | 0          | 0       |
| 2025/03/12 | acbdefghij | 107100003 | C   | 23352      | 1196190 | 51.22        | 1184207    | 50.71           | 0.98           | 0        | 0          | 0       |
| 2025/03/11 | acbdefghij | 107100003 | C   | 13148      | 673515  | 51.22        | 637292     | 48.47           | 0.94           | 0        | 0          | 0       |

### Solution:

```
ALTER SYSTEM ADD STATEMENT HINT ( NO_USE_HEX_PLAN ) COMMENT 'FIX_IT' FOR STATEMENT HASH 'zf2f828810342e6318s875f9c289ceb8';
```

# Workload Management



## Database Hints Tipp

Hints are a very flexible tool and can even be used to assign Workload Classes:

### Example:

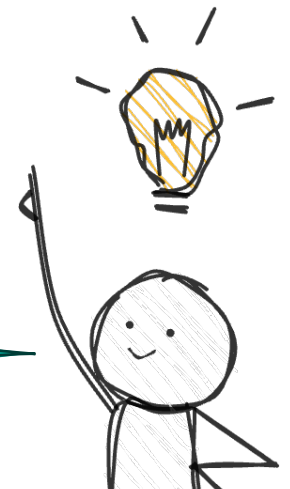
- `CREATE WORKLOAD CLASS "ABC" SET 'TOTAL STATEMENT MEMORY LIMIT' = '1000',  
'TOTAL THREAD LIMIT' = '100';`
- `CREATE WORKLOAD MAPPING "APP_ABC" WORKLOAD CLASS "ABC" SET  
'USER NAME' = 'APP_USER';`

This Restriction for the user can be overruled for a specific statement

```
SELECT * FROM MY_APP WITH HINT( WORKLOAD_CLASS("XYZ") );
```

### Restriction:

Only workload classes which have more restricted properties compared to the ones set for session-wise admission control are applied



# Workload Management

## ➤ Best Practice



### General

Maintain also QAS systems to avoid different behaviour in test cases



### Workload Classes

- A naming concept is strongly recommended
- Workload Classes support hierarchies use them!
- Separation of Workload Classes in categories
  - Business Critical > Important > Monitoring > 3<sup>rd</sup> Party Tools



### Admission Control

- Better to have it set up and don't need it then the other way



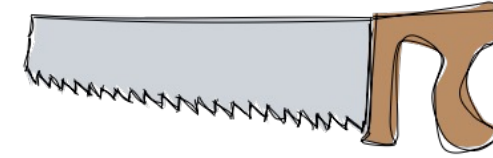
### Hints

- Implement statement hints of [2700051](#) and update them frequently

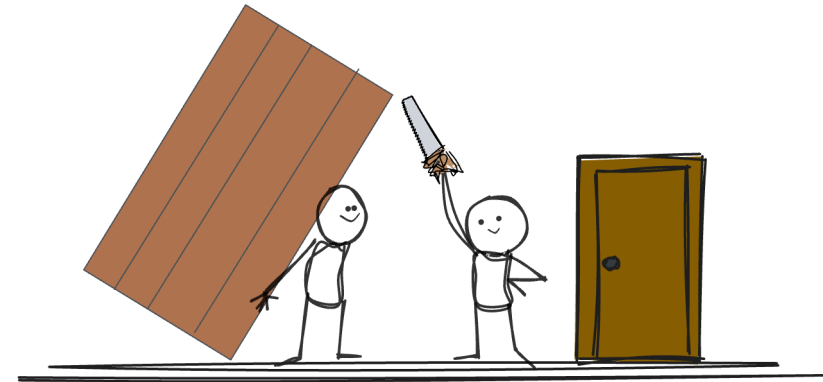


# Overview

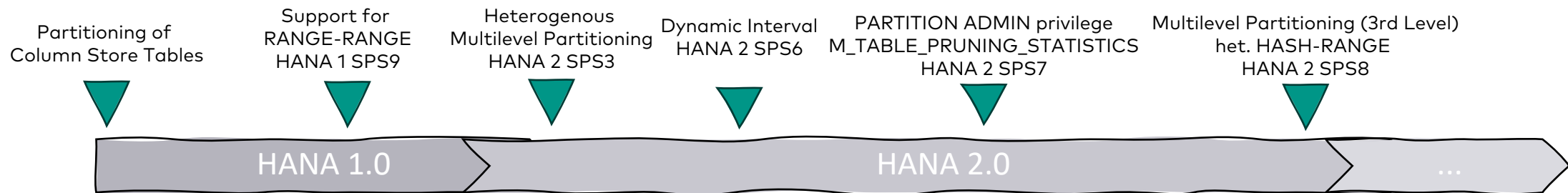
## Partitioning



- Partitioning in SAP HANA is a critical feature for managing **large** datasets in HANA
  - Overcoming Table Size Limits (2B rows/table)
  - Improved Query Performance
  - Enhanced Delta Merge Performance
  - Data Tiering (NSE)
  - Load Balancing in Distributed Systems

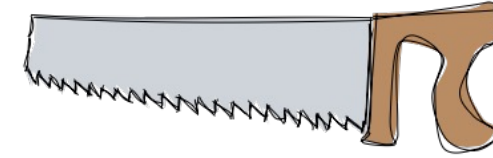


- Over the last SPS new features and functionalities were added as well as steady improved



# Overview

## ➤ Partitioning



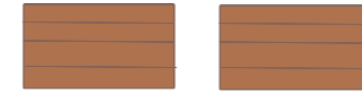
Partitioning helps to ensure performance, scalability and stability by breaking down large tables in better manageable smaller parts.



HANA Table



Partitioning



Partitioned Table

### Performance

- Partition Pruning

### Scalability

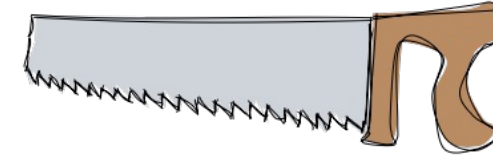
- Load Balancing
- Avoid HANA Limitations

### Stability

- More efficient Merges
- Lower Memory Peaks

# Partitioning

## > Types



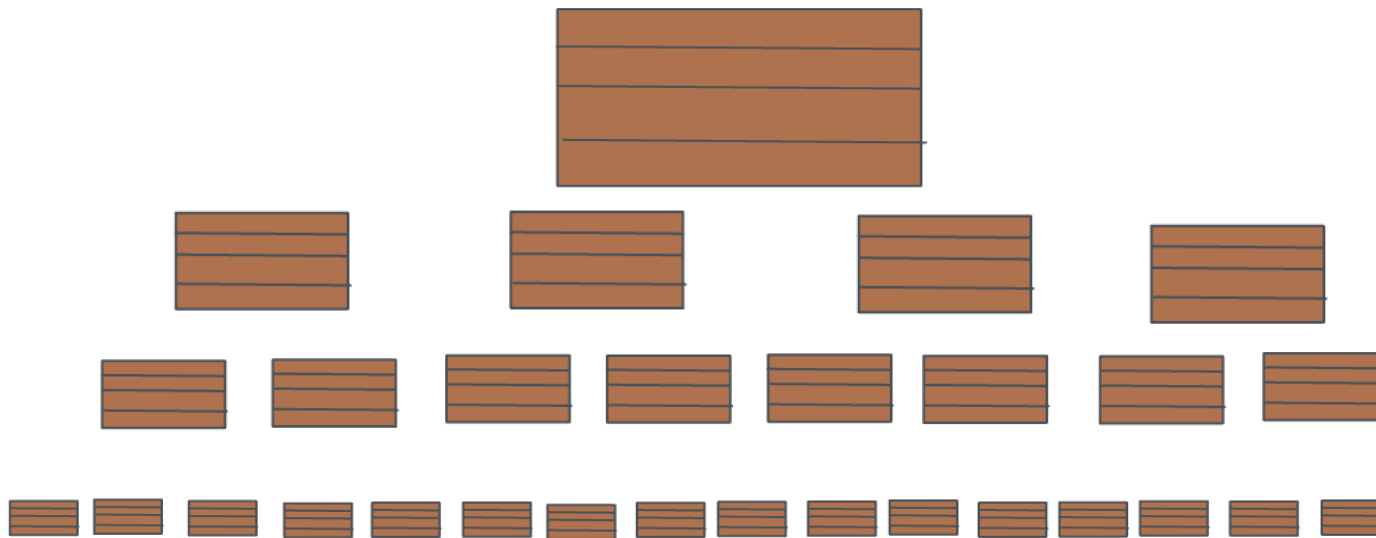
| Types      | Records are assigned...                                                           |
|------------|-----------------------------------------------------------------------------------|
| HASH       | to partitions based on a hash algorithm on the partition key columns              |
| RANGE      | to partitions using defined (non-overlapping) partition ranges                    |
| ROUNDROBIN | in a round-robin manner to the partitions, partition key columns are not required |

Table

Level 1

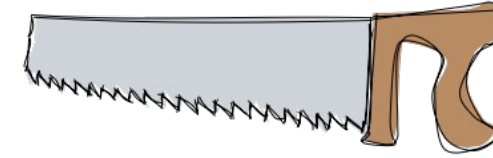
Level 2

Level 3  
(SPS08+)



# Partitioning

## ➤ Partitioning specifications



HO=Homogeneous  
HE=Heterogeneous

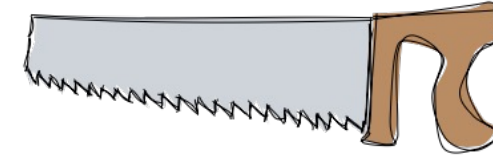
| Partitioning specifications<br>(single- and multi-level) | Homogeneous | Heterogeneous | Only Column of<br>Primary Key                      | NSE support<br>Partition   |
|----------------------------------------------------------|-------------|---------------|----------------------------------------------------|----------------------------|
| HASH                                                     | x           |               | Yes                                                | No                         |
| HASH-HASH                                                | x           |               | Yes                                                | No                         |
| HASH-RANGE                                               | x           | x (=> SPS08)  | HO Level1: Yes<br>HO Level2: No<br>HE Level1/2: No | Yes                        |
| ROUNDROBIN                                               | x           |               | No primary key must<br>be present                  | No                         |
| ROUNDROBIN-RANGE                                         | x           |               | No primary key must<br>be present                  | No                         |
| RANGE                                                    | x           | x             | HO: Yes<br>HE: No                                  | HO: <b>No</b><br>HE: Yes   |
| RANGE-RANGE                                              | x           | x             | HO Level1: Yes<br>HO Level2: No<br>HE Level1/2: No | HO: <b>No</b><br>HE: Yes   |
| RANGE-HASH                                               |             | x             | No                                                 | Yes                        |
| RANGE-RANGE-HASH (=> SPS08)                              |             | x             | No                                                 | Yes (not at HASH<br>level) |
| RANGE-RANGE-RANGE (=> SPS08)                             |             | x             | No                                                 | Yes                        |





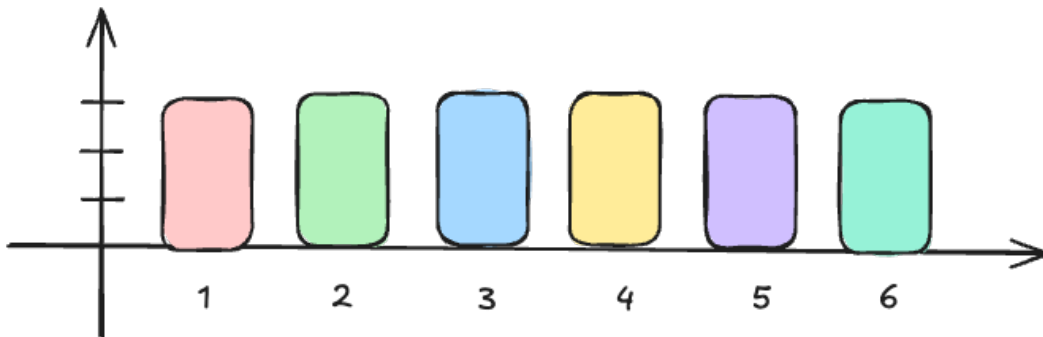
# Partitioning

## ➤ Multilevel Partitioning Variants



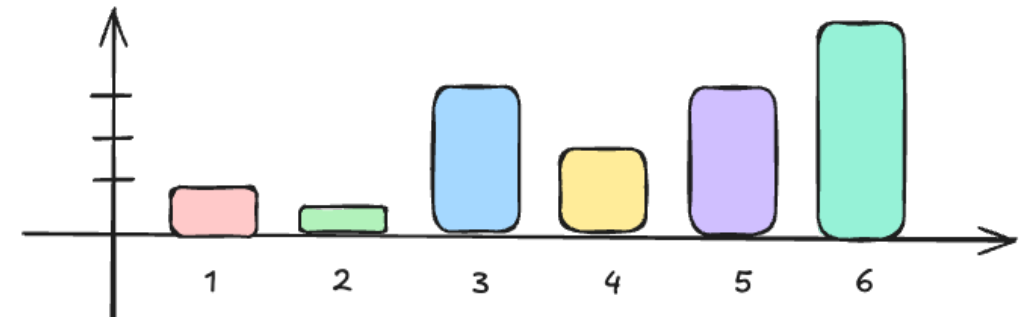
### homogeneous Partitioning

- divides a table into partitions using a **uniform, fixed scheme** on the 2<sup>nd</sup> Level
- The main objective is to achieve an even distribution of data
- Homogeneous partitioning is suitable when the data distribution **is even**
- **Exceptions possible (time-based data)**



### heterogeneous Partitioning

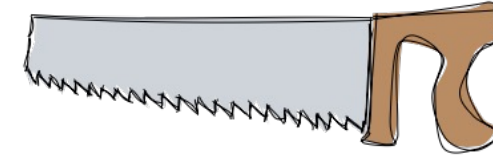
- allows partitions to be defined with **different, specific value ranges** on the 2<sup>nd</sup> Level
- This means that partitions can vary in size and are tailored to the actual data distribution
- Heterogeneous partitioning should be used when data **is not evenly distributed**



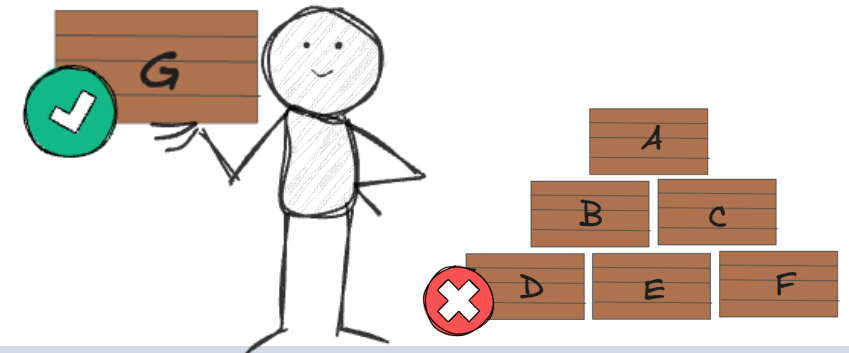
# Partitioning



## Partitioning Considerations

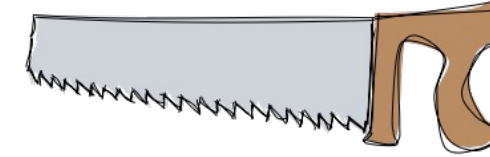


- Difference between Standard Partitioning and Online Partitioning (Details in [2044468](#) )
  - Technically, a different approach to copy the data
  - Different parameters to adjust the Partitioning process
  - Usually, the Standard Partitioning has shorter runtime compared to Online Partitioning at the cost of more resource consumption
- Partition Pruning
  - Partition Pruning is an efficient way to provide a performance boost in large HANA Systems
  - The intention is to just read the relevant partitions instead of all partitions of the table
  - The Most important SQL Statements should use the selected column for the partitioning as filter criteria



# Partitioning

## Partitioning Pruning Example



- RANGE Partitioning on Column DEP
  - SELECT DATA WHERE DEP = 'IT'
  - SELECT DATA WHERE NAME = 'MARTIN'

| PNR  | NAME   | DEP   | DATA              |
|------|--------|-------|-------------------|
| 7369 | SMITH  | HR    | AFBIDABVUIDABVL   |
| 7499 | ALLEN  | HR    | AISOHAIOVHIOADVH  |
| 7521 | WARD   | HR    | DIBHIUDFBHDBDB    |
| 7566 | JONES  | HR    | EHG,XEHUHQBV=DEH  |
| 7654 | MARTIN | IT    | SOICAHSOICHVDVW   |
| 7698 | BLAKE  | IT    | SVIAIOUVBUBVOB    |
| 7782 | CLARK  | SALES | IOAHVOIHASVOH     |
| 7788 | SCOTT  | SALES | AOIVBODVBAOIDVB   |
| 7839 | KING   | SALES | ATSH)HQQ,AUSGFS(F |
| 7844 | TURNER | PROD  | JVBIAUSDVKEHFISO  |
| 7876 | ADAMS  | PROD  | ADNVOIAHDVHASH    |
| 7900 | JAMES  | PROD  | HAIUSHASHVASVA    |
| 7902 | FORD   | PROD  | WSAOIHVASH=A)SFH  |
| 7934 | MILLER | PROD  | IOAVHD)HVASHJDQ   |

RANGE  
Partition by  
"DEP"

| PNR  | NAME   | DEP   | DATA              |
|------|--------|-------|-------------------|
| 7369 | SMITH  | HR    | AFBIDABVUIDABVL   |
| 7499 | ALLEN  | HR    | AISOHAIOVHIOADVH  |
| 7521 | WARD   | HR    | DIBHIUDFBHDBDB    |
| 7566 | JONES  | HR    | EHG,XEHUHQBV=DEH  |
| 7654 | MARTIN | IT    | SOICAHSOICHVDVW   |
| 7698 | BLAKE  | IT    | SVIAIOUVBUBVOB    |
| 7782 | CLARK  | SALES | IOAHVOIHASVOH     |
| 7788 | SCOTT  | SALES | AOIVBODVBAOIDVB   |
| 7839 | KING   | SALES | ATSH)HQQ,AUSGFS(F |
| 7844 | TURNER | PROD  | JVBIAUSDVKEHFISO  |
| 7876 | ADAMS  | PROD  | ADNVOIAHDVHASH    |
| 7900 | JAMES  | PROD  | HAIUSHASHVASVA    |
| 7902 | FORD   | PROD  | WSAOIHVASH=A)SFH  |
| 7934 | MILLER | PROD  | IOAVHD)HVASHJDQ   |

SELECT  
DATA  
WHERE  
DEP = 'IT'

SELECT  
DATA  
WHERE  
NAME =  
'MARTIN'

| PNR  | NAME   | DEP   | DATA              |
|------|--------|-------|-------------------|
| 7369 | SMITH  | HR    | AFBIDABVUIDABVL   |
| 7499 | ALLEN  | HR    | AISOHAIOVHIOADVH  |
| 7521 | WARD   | HR    | DIBHIUDFBHDBDB    |
| 7566 | JONES  | HR    | EHG,XEHUHQBV=DEH  |
| 7654 | MARTIN | IT    | SOICAHSOICHVDVW   |
| 7698 | BLAKE  | IT    | SVIAIOUVBUBVOB    |
| 7782 | CLARK  | SALES | IOAHVOIHASVOH     |
| 7788 | SCOTT  | SALES | AOIVBODVBAOIDVB   |
| 7839 | KING   | SALES | ATSH)HQQ,AUSGFS(F |
| 7844 | TURNER | PROD  | JVBIAUSDVKEHFISO  |
| 7876 | ADAMS  | PROD  | ADNVOIAHDVHASH    |
| 7900 | JAMES  | PROD  | HAIUSHASHVASVA    |
| 7902 | FORD   | PROD  | WSAOIHVASH=A)SFH  |
| 7934 | MILLER | PROD  | IOAVHD)HVASHJDQ   |



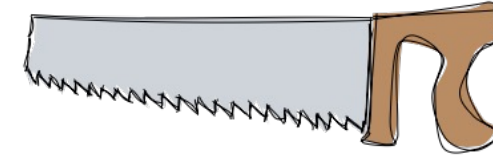
| PNR  | NAME   | DEP   | DATA              |
|------|--------|-------|-------------------|
| 7369 | SMITH  | HR    | AFBIDABVUIDABVL   |
| 7499 | ALLEN  | HR    | AISOHAIOVHIOADVH  |
| 7521 | WARD   | HR    | DIBHIUDFBHDBDB    |
| 7566 | JONES  | HR    | EHG,XEHUHQBV=DEH  |
| 7654 | MARTIN | IT    | SOICAHSOICHVDVW   |
| 7698 | BLAKE  | IT    | SVIAIOUVBUBVOB    |
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| 7902 | FORD   | PROD  | WSAOIHVASH=A)SFH  |
| 7934 | MILLER | PROD  | IOAVHD)HVASHJDQ   |



# Partitioning



## Execution Tipps (Standard)



1. Define Parameters for Partitioning based on available Ressources:
  - `[partitioning], 'split_threads'`
  - `[partitioning], 'bulk_load_threads'` (only available until SPS08)
  - `[indexing], 'parallel_merge_threads'`
2. Disable OPTIMIZE COMPRESSION (optional)
  - `ALTER TABLE "SCHEMA"."TABLE" WITH PARAMETERS ('AUTO_OPTIMIZE_COMPRESSION' = 'OFF');`
3. Load Table
  - `LOAD "SCHEMA"."TABLE" ALL;`
4. Delta Merge – wait until the merge is finished
  - `MERGE DELTA OF "TABLE";`
5. Drop Primary Key (optional)
  - `ALTER TABLE "SCHEMA"."TABLE" DROP PRIMARY KEY;`
6. Deactivation of User Limits
  - `alter user <Username> set parameter statement memory limit = '0';`
  - `alter user <Username> set parameter statement thread limit = '0';`
7. Start of Partitioning
  - `ALTER TABLE "SCHEMA"."TABLE" PARTITION BY ...`

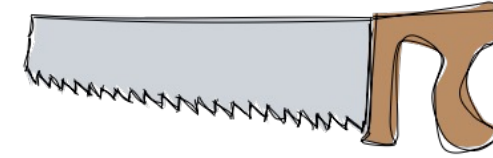
Don't forget to revert the optional steps 2., 5. and 6. afterwards!



# Partitioning



## Execution Tipps (Online)



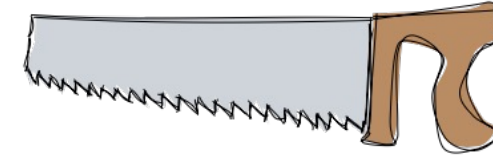
### Parameter Recommendation Note by SAP:

- [2874176](#) - Parameter Recommendations for Online Table Operations (Legacy note for SPS05)
- [3469162](#) - Online Partitioning SAP HANA 2 SPS07
- No specific note for SPS08 available

1. Set the parameters according to the note that fits for your support package
2. Disable OPTIMIZE COMPRESSION (optional)
  - `ALTER TABLE "SCHEMA"."TABLE" WITH PARAMETERS ('AUTO_OPTIMIZE_COMPRESSION' = 'OFF')`
3. Load Table
  - `LOAD "SCHEMA"."TABLE" ALL;`
4. Optimize Delta Merge
  - `[mergedog], 'auto_merge_decision_func'`
5. Deactivation of User Limits
  - `alter user <Username> set parameter statement memory limit = '0';`
  - `alter user <Username> set parameter statement thread limit = '0';`
6. Start of Partitioning
  - `ALTER TABLE "SCHEMA"."TABLE" PARTITION BY ... ONLINE`

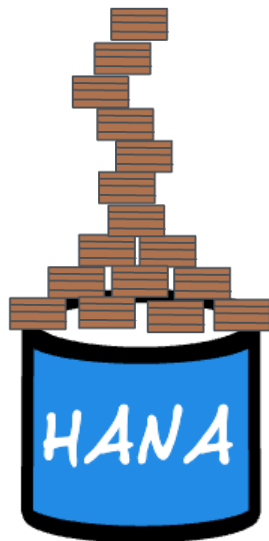


# Partitioning Example

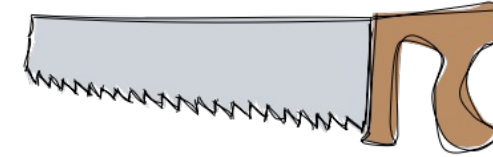


## ➤ Partitioning Example CAR system

- Effected tables /POSDW/TLOGF and /POSDW/TLOGF\_EXT
  - First Level HASH ("BUSINESSDAYDATE","RETAILSTOREID") PARTITIONS 4
  - Second Level: RANGE ("BUSINESSDAYDATE") (PARTITION '20250601' <= VALUES < '20250608' PAGE LOADABLE, and many more for every week.
- Due to the large amount of data and a country that is not allowed to be archived, we ended up with around 140 partitions per table
  - Which is technically not a problem. The limit is 16000 (Source: [2154870](#))
  - Note: You could change parameter *compile\_time\_sampling\_partitions* if you have such a high number of partitions (Be aware that higher values result in better estimations, but they can increase the parse times, so adjustments need to be done with care. Source: [2000002](#))



# Partitioning Example



## Partitioning Example CAR system

- After the Go Live of the partition design a new Z-program was developed and set live

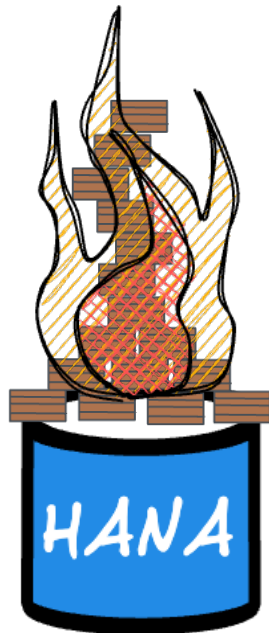
|   | OPERATOR_NAME ▾ | OPERATOR_DETAILS ▾                                                       |
|---|-----------------|--------------------------------------------------------------------------|
| 1 | COLUMN SEARCH   | T1.MANDT, T1.RETAILSTOREID, T1.BUSINESSDAYDATE                           |
| 2 | JOIN            | JOIN CONDITION: (INNER) TX.BUSINESSDAYDATE = T1.BUSINESSDAYDATE          |
| 3 | COLUMN TABLE    | FILTER CONDITION: TX.MANDT = n'701' AND TX.RETAILSTOREID = n'0000000017' |
| 4 | COLUMN TABLE    | FILTER CONDITION: T1.MANDT = n'701' AND T1.RETAILSTOREID = n'0000000017' |

```
SELECT
T1.MANDT,
T1.RETAILSTOREID,
T1.BUSINESSDAYDATE
FROM "SCHEMA"."TEST_PRUNGING" as TX
INNER JOIN "SCHEMA"."TEST_PRUNGING1" as T1
ON TX.MANDT = T1.MANDT
AND TX.RETAILSTOREID = T1.RETAILSTOREID
AND TX.BUSINESSDAYDATE = T1.BUSINESSDAYDATE
AND TX.MANDT = '701'
AND TX.RETAILSTOREID = '0000000017'
```

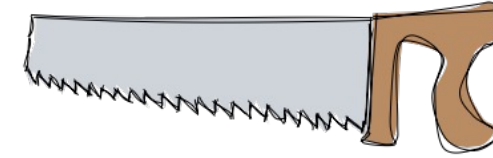


### Problems

- ✗ BUSINESSDAYDATE and RETAILSTOREID are materialized and used for the JOIN, but not in the WHERE CLAUSE
- ✗ No partition pruning is performed
- ✗ Round about 280 partitions must be processed
- ✗ Every night there was a over 90% CPU situation -> Admission Control was active



# Partitioning Example



## Partitioning Example CAR system

- The best solution would be to include the BUSINESSDAYDATE in the WHERE-clause

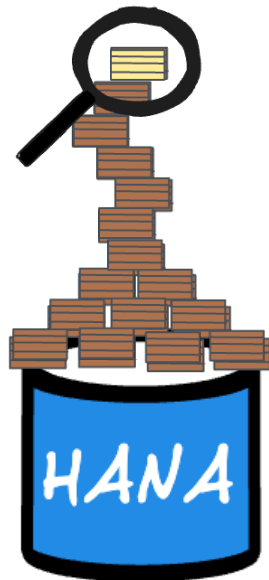
|   | OPERATOR_NAME ▾ | OPERATOR_DETAILS ▾                                                                                                                   |
|---|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1 | COLUMN SEARCH   | T1.MANDT, T1.RETAILSTOREID, T1.BUSINESSDAYDATE                                                                                       |
| 2 | JOIN            | JOIN CONDITION: (INNER) TEST_PRUNGING x TEST_PRUNGING1                                                                               |
| 3 | COLUMN TABLE    | USED PARTITION IDS: 34 FILTER CONDITION: TX.MANDT = n'701' AND TX.RETAILSTOREID = n'0000000017' AND TX.BUSINESSDAYDATE = n'20220610' |
| 4 | COLUMN TABLE    | USED PARTITION IDS: 34 FILTER CONDITION: T1.MANDT = n'701' AND T1.RETAILSTOREID = n'0000000017' AND T1.BUSINESSDAYDATE = n'20220610' |

```
SELECT
T1.MANDT,
T1.RETAILSTOREID,
T1.BUSINESSDAYDATE
FROM "SCHEMA"."TEST_PRUNGING" as TX
INNER JOIN "SCHEMA"."TEST_PRUNGING1" as T1
ON TX.MANDT = T1.MANDT
AND TX.RETAILSTOREID = T1.RETAILSTOREID
AND TX.BUSINESSDAYDATE = T1.BUSINESSDAYDATE
AND TX.MANDT = '701'
AND TX.RETAILSTOREID = '0000000017'
AND TX.BUSINESSDAYDATE = '20250610'
```



### Problems

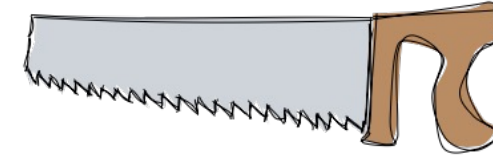
- ✓ BUSINESSDAYDATE and RETAILSTOREID are materialized and used for the JOIN, and in the WHERE CLAUSE
- ✓ partition pruning is performed
- ✓ Just 1 partition must be processed
- ✓ CPU Load would be reduced
- ✗ However, this was not possible in the customer's case ...





# Partitioning

## ➤ Best Practise



### Partition sizes

- 250-500 Mio. Records
  - There might be some exceptions for specific tables, like ACDOCA
- 25-50 GB
  - If you are using NSE the partitions must fit into the buffer cache

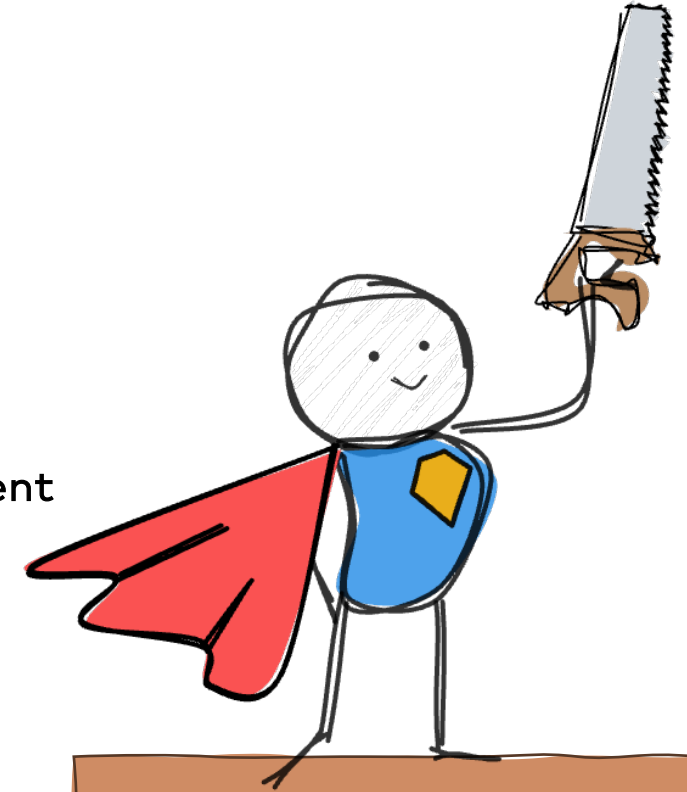
### Write down all partitioned tables and share the information with development

- Partition Pruning!

### Test your design before using it in a productive system

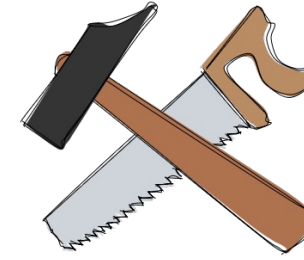
- Try to observe all use cases throughout the year

### Consider NSE in the partition design if it is already planned



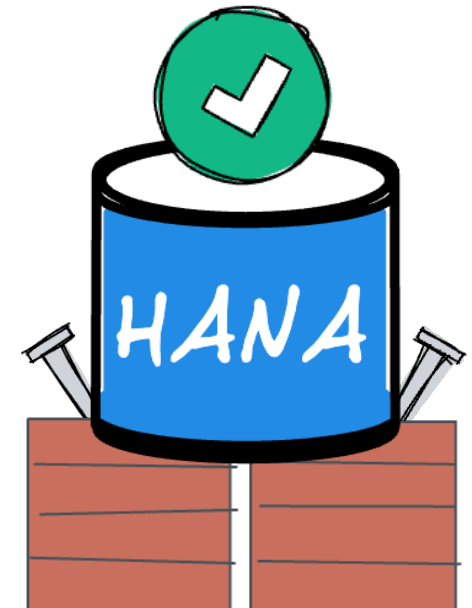
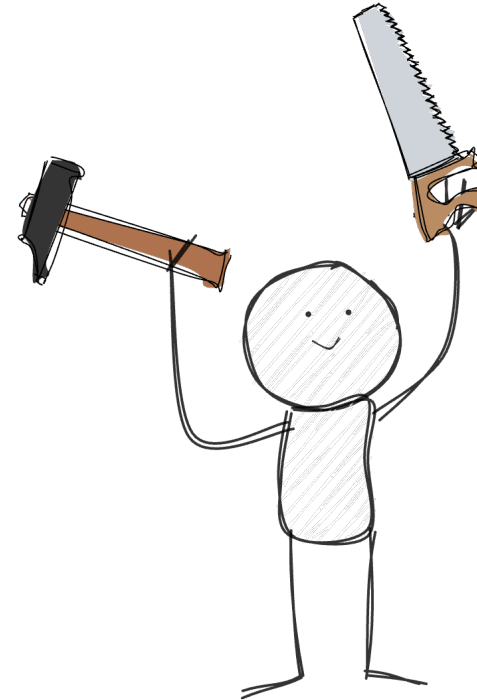
[2044468](#) - FAQ: SAP HANA Partitioning

# Combined Use Cases



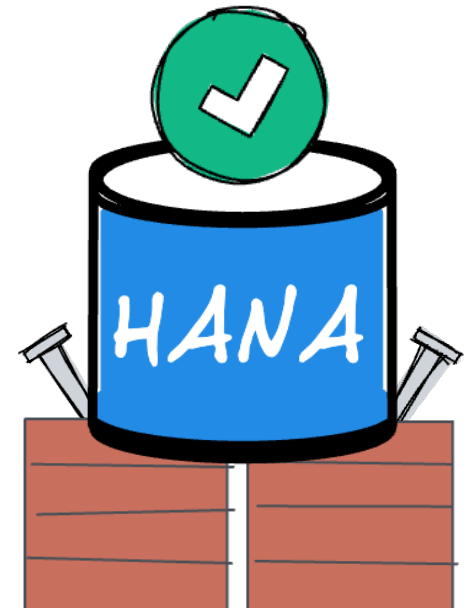
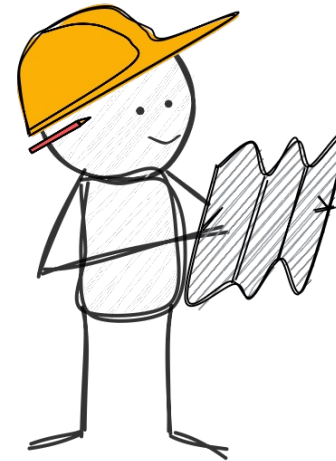
## ➤ Workaround for CAR system partition problem

- During a transition phase, the Z-program must run every night
- **Workaround:**
  - Let the statement hash run in specific workload class
  - The Z program needs more time, but the database no longer activates Admission Control
  - All other users can work normally



# Summary

- Both tools are useful additions to the daily life of a HANA admin
- They should not be missing in any SAP HANA toolbox



# More Deep Dive ...

... Examples and additional Information  
to this and other Topics can be found in  
the first Book written by

Matthias Sander

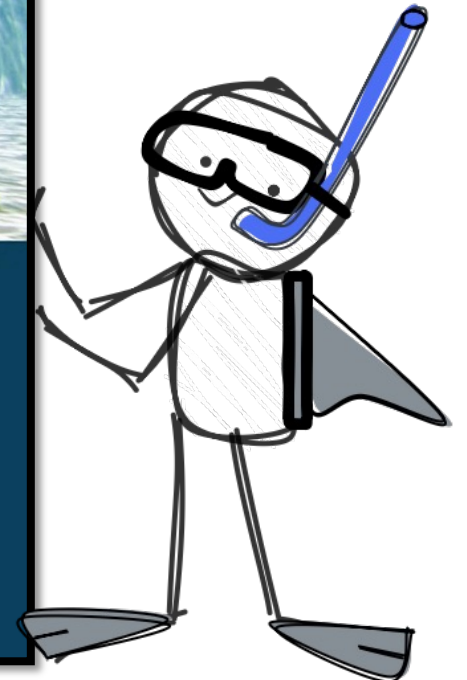
and

Jens Gleichmann

ISBN: 978-3-960124-18-4

(currently only in german)

...but a english Version will follow soon





# Thank you for your attention

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